

Progressive Pull Request — SMP integration

**What, why, file and code, from the corrections to the SMP
integration**

Progressive Pull Request — SMP integration into xpack-development

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Progressive Pull Request — SMP integration

into `xpack-development`

What to change, why, in which file, and with which code — step by step, from the corrections to the SMP integration.

This document turns the internal integration runbook (`Implementation-SMP-Integration.md`) and the per-file inventory (`files-modif-by-step.md`) into a **progressive pull request**: one reviewable change after another, starting with the single-core corrections and ending with the multi-core (SMP) integration.

Everything below refers to the same model:

- `smp` — the branch that already contains the finished work (source of truth, never modified).
- `xpack-development` — the destination branch the work is rebuilt onto, one cohesive step at a time.
- Every step is **one commit** (tagged `step-NN-green`) and leaves the tree **green**: the frozen 72-test suite still passes, and for the SMP part the single-core projection is byte-identical to the previous step.

Table key. Repo : `K` = `micro-os-plus-iii` (kernel) · `C` = `micro-os-plus-iii-cortexm` · `P` = `micro-os-plus-iii-posix-arch` . **How:** `whole` = `git checkout origin/smp -- <file>` ; `projection` = `git show origin/smp:<file> | scripts/smp/sc-project.py > <file>` (strips the `OS_USE_SMP_SCHEDULER` / `NCPU>1` blocks); `surgical` = insert only the named hunk with `scripts/smp/chunks/_edit.py` . The executable form of every row is `scripts/smp/chunks/stepNN.sh` .

Synchronised with the branches. This plan was checked against the actual `smp` and `xpack-development` trees and the recipes in `scripts/smp/chunks/` : every file listed appears in the `xpack-development..smp` diff, and the lift method matches the recipe (`whole` / `sc-project.py` / `_edit.py`). Three runbook realities are reflected: Part B is **collapsed into a single `step/14` commit** (steps 15–23 never get their own branch); `tests/smp-support/` , `native-smp` and the `test-smp-all` action are **created by the integration**, not present on `smp` ; and the `src/libc/stdlib/timemgm.c` change is deliberately **not** lifted (the baseline version is kept).

1. The pull request at a glance

Field	Value
Title	SMP multi-core integration (progressive, bisectable series)
Repositories	<code>micro-os-plus-iii</code> · <code>-cortexm</code> · <code>-posix-arch</code> (and the add-only arch repos)
Base branch	<code>xpack-development</code>
Head branch	<code>step/30</code> (or the stacked <code>step/NN</code> chain)
Source branch	<code>smp</code> (read-only)
Scope	Part 0 → Part A (corrections) → Part B (SMP) → Part C (targets) → Part D (docs)
Commits	one per step (Part B is one collapsed <code>step/14</code> commit), each tagged <code>step-NN-green</code>
Acceptance gate	pristine check + 72/72 tests + <code>unifdef</code> invariant (Part B) + <code>test-smp-all</code> (Part C)
Push policy	nothing is pushed until the PR is opened by hand

1.1 How the progressive PR is ordered

The order is **corrections** → **SMP evolution** → **SMP tests** → **new targets/tests** → **final integration**, with the two infrastructure items (the `devices` dissolution and the port releases) placed where their dependencies actually are, not first.

PR	Contents	Steps	Repos	Depends on
#1	Corrections (single-core correctness)	Part A (1–13)	K, +C+P at 13	—
#2	SMP evolution (collapsed <code>step/14</code>) + the genuine <code>native-smp</code> dual-core platform, its configs and <code>test-native-smp</code>	Part B (14–23)	K, C, P	#1
#3	Port releases: <code>posix-arch v1.1.0</code> , <code>cortexm v1.2.0</code>	24	P, C	#2
#4	Dissolve the <code>devices</code> repository	Part 0	P, C, A32, A64	#3
#5	New cores/boards + modular CMake	25–26	K, C, P, arch	#2, #4
#6	New test sources + add-only platforms (+ <code>test-smp-all</code>)	27–28	K, C, P, arch	#5
#7	Documentation + final merge	29–30	all	#1–#6

PR head branches: `step/13` → `step/14` → `step/24` → `step/26` → `step/28` → `step/30` (the intermediate `step/25` and `step/27` are commits inside PRs #5 and

6). Part 0 (#4) is out of band on the port and architecture repos only — it

never touches the kernel — so it is applied on its own `part0-devices` branch before #5 and is not in the kernel's linear chain. Part B is collapsed, so `step/15` ... `step/23` do not exist.

Two equivalent ways to use this:

- **One progressive PR** (recommended for a single reviewer): open #1, and add the later parts as new commits on the same head as they are accepted.
- **Stacked PRs** (recommended for a large review): open #1, then #2 against `step/13`, and so on; each merges into the one below it.

1.2 Order verification — is this optimal?

The phase order is forced by the dependency graph. Each claim below was checked against the code, not by analogy.

Question	Evidence	Verdict
Must the corrections come first?	Part B's Steps 9/10/13 build on Part A's <code>destroying</code> state, condvar <code>clock</code> and highres-clock port sync; Part A is valuable on its own and shrinks the SMP diff.	Yes — first.
Does Part B need Part 0 (dissolve <code>devices</code>)?	Neither port's <code>xpack-development</code> <code>CMakeLists.txt</code> references <code>devices</code> , and <code>part-b.sh</code> fix C deliberately does not apply <code>smp</code> 's <code>UOS_DEVICES_DIR</code> CMake.	No. Part 0 only unblocks the new boards, so it moves after Part B.
Does Part B have its own test?	<code>verify-step.sh</code> Stage 4 is only the NCPU=2 smoke, which the runbook documents as fake (a CMake cache variable never becomes a compile definition); the real <code>native-smp</code> / <code>2xcortex-m33</code> tests only arrive in Part C.	No — the real gap. Ship <code>native-smp</code> with Part B.
Do the new cores/boards need Part B?	The M33/RP2350 cores implement the SMP port contract introduced in Part B.	Yes — after Part B.
Do the new boards need Part 0?	<code>soc/rp2350</code> , <code>soc/stm32f4xx</code> , <code>soc/bcm2837</code> and the shared drivers come from <code>devices</code> .	Yes — Part 0 before the new boards.
Do the tests need the new cores/boards?	<code>2xcortex-m33</code> needs the M33 port (Step 25) and the modular CMake (Step 26).	Yes — tests after evolutions.
Is the final merge last?	Step 30 reconciles with <code>xpack-development</code> and restores <code>.github/**</code> .	Yes — last.

The original packaging had three non-optimal points, now corrected: Part 0 was first (it is independent of Part A/B), the port releases were buried inside the tests PR (they publish the SMP evolution), and **Part B shipped without a genuine dual-core test** (its only SMP check was the documented-fake smoke). The new `native-smp` leg fixes the last one by landing the real dual-core test with the code it tests.

1.3 Is each step self-contained?

Short answer: **each PR is self-contained; each individual step is a self-contained *delta*, not a self-contained *feature*.** A step is green given all the steps before it, but it is not meant to be applied to the baseline on its own. Two forces break standalone independence — **atomic cross-repo/definition steps** and **sequential in-file dependencies** — and Part B is not separable at all.

Unit	Self-contained?	Why (evidence)
Each PR (#1–#7)	Yes , given its base	<code>verify-step.sh</code> / <code>full-verify.sh</code> gate it; it builds and passes on the branch it targets.
Part A Steps 1–12	Delta only	Applied in order; a later step may use an earlier symbol (e.g. Step 9's <code>destroying</code> state).
Step 13	Delta + atomic across K, C, P	K declares <code>has_hardware_counter()</code> , C/P define it; landing only the kernel breaks the link (the runbook's <code>qemu-cortex-m7f-release</code> failure).
Part B (14–23)	Not separable	One interwoven <code>#if defined(OS_USE_SMP_SCHEDULER)</code> blob; <code>part-b.sh</code> states "not per-step bisectable by design"; collapsed into one <code>step/14</code> commit.
Step 24	Yes, given Part B	Version-only change (<code>package.json</code> <code>version</code>).
Step 25	Yes, given Part B + Part 0	New port files; atomic across C/P.
Step 26	Needs 25	The M33 platform links the M33 port target.
Step 27	Needs 26	The new suites are wired into the new M33 platform.
Step 28	Needs 26–27	The AArch platforms use the harness and tests added before.
Part 0	Yes (port/arch only)	Independent of the kernel; its gate is "boards that linked <code>devices</code> still build".

Why some steps cannot stand alone:

- **Atomic definition + declaration.** Step 13 adds `port::clock_highres:: has_hardware_counter()` in the kernel and its definition in the two ports; the two halves must be committed together or the link fails. Part B (Step 14) and Step 25 are the same: the kernel's `port_cpu_id()` and the port's definition, or the M33 core and its CMake target, travel together.
- **In-file chaining.** `os-c-wrapper.cpp` is touched by Steps 4, 9 and 20; `os-thread.{h,cpp}` by 1, 7, 9, 14 and 20; `os-c-decls.h` by 9, 10 and 16; `os-core.cpp` by 14–22. The recipes use **whole / projection** / **surgical** lifting so each step carries only *its* hunks — but it still assumes the earlier steps are already in place (see §11.2).
- **Deliberate coupling.** Step 4 strips the Step-9 `destroying` `static_assert` because it does not compile yet; Step 7 keeps Step-9/20 content out to avoid `-Werror=switch-enum`. The steps are curated to *avoid* pulling the future in, not to make each one buildable from the baseline.

The rule the recipes enforce: a step may lift a **partial file**, but never a **partial dependency** — a declaration and its definition, or a test and the platform it runs on, always travel in the same step (or PR).

2. Phase 0 — Dissolve the `devices` repository (PR #4)

Why the whole part. `xpack-development` has six components; `smp` adds a seventh, `micro-os-plus-iii-devices`. Upstream has no such repo and nowhere to put one. Folding its code (history-preserving `git subtree`) into the architecture repos that already own it lets the rest of the integration proceed with the six repos upstream has.

Repo	What	Why	File / path	How (code)
P	Move the native SoC	host SD-image backend belongs to the host port	<code>soc/native/</code>	<code>git subtree add -P soc/native <devices> split/soc_native</code>
P	Move the shared drivers	block/FatFs/USB drivers are portable C+++, used by boards	<code>drivers/include/</code> , <code>drivers/src/</code> , <code>drivers/fatfs/</code>	<code>git subtree add -P drivers/... <devices> split/...</code>
P	Carry driver fixes	atomic-spinlock mailbox; flatfs append-to-empty allocates an extent	<code>src/.../mailbox</code> , <code>fatfs</code>	part of the subtree move
C	Move the STM32F4xx SoC	Cortex-M silicon owned by the Cortex-M port	<code>soc/stm32f4xx/</code>	<code>git subtree add -P soc/stm32f4xx <devices> split/soc_stm32f4xx</code>
C	Move the RP2350 SoC	Pico 2 headers + <code>system_rp2350.c</code>	<code>soc/rp2350/</code>	<code>git subtree add -P soc/rp2350 <devices> split/soc_rp2350</code>
A32/A64	Move drivers + <code>bcm2837</code>	each arch repo must be self-contained (no cross-repo dep)	<code>drivers/</code> , <code>soc/bcm2837/</code>	<code>git subtree add ...</code>
P, C, A32, A64	Re-declare the old CMake target names	so every board that linked <code>micro-os-plus::devices</code> keeps working	<code>CMakeLists.txt</code> (shim)	<code>add_library(... INTERFACE) + add_library(micro-os-plus::devices ALIAS ...)</code>

Gate for PR #4: every board that linked a `devices` target still configures and builds; the standalone `devices` repo is no longer a dependency.

3. Phase A — Corrections: single-core correctness (PR #1, Steps 1–13)

Why the whole part. These fixes are correct and valuable **without** SMP. Landing them first keeps each step bisectable and makes Part B a pure SMP diff.

3.1 Steps

Step	What	Why	File	Code / how	Gate
1	ISO C conformance, list iterators, exported suspend, newlib return type	host glibc double-declares <code>DIR</code> ; intrusive-list iterators must conform; the C wrapper must resolve <code>thread::suspend</code> ; newlib's <code>read / write</code> return type must match the weak alias	<code>include/cmsis-plus/posix/dirent.h</code> , <code>include/cmsis-plus/utils/lists.h</code> , <code>include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h</code> , <code>src/rtos/os-thread.cpp</code>	whole for the three headers — <code>c-syscalls-aliases-standard.h</code> defines <code>CMSIS_PLUS_POSIX_IO_RW_RETURN_TYPE</code> as newlib's <code>_READ_WRITE_RETURN_TYPE</code> (or <code>ssize_t</code>); surgical: remove <code>inline</code> from <code>this_thread::suspend</code> in <code>os-thread.cpp</code> . Keep baseline <code>src/libc/stdlib/timegm.c</code>	72/72
2	Memory: overflow, usable size, calloc/ realloc, block-pool	header decl + base definition + override must land together or <code>memory_resource::do_usable_size</code> is undefined; overflow guards	<code>include/cmsis-plus/rtos/os-memory.h</code> , <code>include/cmsis-plus/memory/first-fit-top.h</code> , <code>src/rtos/os-memory.cpp</code> , <code>src/memory/first-fit-top.cpp</code> , <code>src/memory/lifo.cpp</code> , <code>src/memory/block-pool.cpp</code> , <code>src/libc/stdlib/malloc.cpp</code>	whole (7 files); add <code>-Wcast-align / -Wunsafe-buffer-usage</code> pragma wrap in <code>first-fit-top.cpp</code> ; calloc guard <code>if (nelem != 0 && elbytes > (SIZE_MAX / nelem)) { errno = ENOMEM; return nullptr; }</code>	72/72; <code>mutex-stress</code> + allocator paths
3	C++17 aligned new/delete, static error category, chrono	aligned allocation; exit-time-destructor warning; 64-bit overflow	<code>src/libcpp/new.cpp</code> , <code>src/libcpp/system-error.cpp</code> , <code>src/libcpp/chrono.cpp</code>	whole; add clang <code>-Wexit-time-destructors</code> suppression for the two Meyers-singleton categories; <code>(cyc/f)*1e9 + (cyc%f)*1e9/f</code>	72/72
4	C wrapper: one-shot timer, polymorphic	correct timer default; delete the right type; cast to	<code>src/rtos/os-c-wrapper.cpp</code>	whole, then strip Step-9 <code>destroying</code>	72/72

Step	What	Why	File	Code / how	Gate
	delete, 64-bit timeouts	64-bit before <code>*</code> <code>1000u</code>		static_assert and Step-20 SMP affinity blocks	
5	POSIX I/O thread-safety	descriptor-table race; free-list race; inverted size check	<code>src/posix-io/file-descriptors-manager.cpp</code> , <code>include/cmsis-plus/posix-io/file-system.h</code> , <code>include/cmsis-plus/posix-io/net-stack.h</code> , <code>src/posix-io/block-device.cpp</code>	whole; add <code>-Wunsafe-buffer-usage</code> suppression in the fd manager; <code>if (size == 0) return EINVAL;</code>	72/72
6	ARMv8-M guards, SecureFault, semihosting fstat	M33 build; fault handler; <code>S_IFCHR</code> only when no type set	<code>include/cmsis-plus/arm/semihosting.h</code> , <code>src/startup/exception-handlers.c</code> , <code>src/semihosting/c-syscalls-semihosting.cpp</code>	whole; add <code>__ARM_ARCH_8M_MAIN__ / __ARM_ARCH_8M_BASE__</code> guards + <code>SecureFault_Handler</code> ; add weak <code>os_board_console_mirror()</code>	72/72
7	Timer callback outside the critical section	avoid running user code while the lock is held; periodic catch-up re-arm	<code>src/rtos/internal/os-lists.cpp</code> , <code>src/rtos/os-timer.cpp</code> , <code>include/cmsis-plus/rtos/os-thread.h</code>	whole for the two sources; surgical errno-scratch hunk into <code>os-thread.h</code> (keep Step-9/20 out)	72/72
8	Mutex priority ceiling	set ceiling before ownership; boost to max waiter; recompute owner priority on unlock	<code>src/rtos/os-mutex.cpp</code>	whole	72/72
9	Thread lifecycle: <code>state::destroying</code> , atomic join, detach, reaper	race-free destruction/join; single-lock register/suspend	<code>include/cmsis-plus/rtos/os-thread.h</code> , <code>src/rtos/os-thread.cpp</code> , <code>src/rtos/os-idle.cpp</code> , <code>include/cmsis-plus/rtos/os-c-decls.h</code>	projection for the three (<code>os-thread.cpp</code> has ~37 SMP hunks); surgical <code>destroying = 7</code> into <code>os-c-decls.h</code> (keep Step-10 <code>void* clock</code> out)	72/72

Step	What	Why	File	Code / how	Gate
10	Condition-variable atomicity	link-then-unlock-then-suspend; bind <code>timed_wait</code> to the attribute clock	<code>src/rtos/os-condvar.cpp</code> , <code>include/cmsis-plus/rtos/os-condvar.h</code> , <code>include/cmsis-plus/diag/instrumentation.h</code> , <code>include/cmsis-plus/rtos/os-c-decls.h</code>	whole for the three; surgical : uncomment only <code>os_condvar_t</code> 's <code>void* clock</code> ; add <code>OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_CONDVAR (12)</code>	72/72
11	<code>std::thread</code> functor lifetime & join	join the native handle before delete; deterministic free	<code>src/libcpp/thread-cpp.h</code> , <code>include/cmsis-plus/estd/thread_internal.h</code>	whole	72/72
12	Message-queue reschedule	preempt after a send/receive wakes a higher-priority thread	<code>src/rtos/os-mqueue.cpp</code>	whole; <code>port::scheduler::reschedule()</code> after the critical section in all 6 paths	72/72
13	High-resolution clock port sync (first cross-repo step: K + C + P)	declare/call <code>has_hardware_counter()</code> ; fix SysTick pending test; read <code>CLOCK_MONOTONIC</code>	K: <code>include/cmsis-plus/rtos/os-decls.h</code> , <code>src/rtos/os-clocks.cpp</code> ; C: <code>include/cmsis-plus/rtos/port/os-inlines.h</code> ; P: same file	K whole; C projection (<code>has_hardware_counter()→false</code> , <code>hardware_counter()→0</code>); P surgical insert of only the <code>clock_highres</code> block	72/72

Step 1 completeness note. The runbook's detailed Step-1 table also lists `include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h`; the current `chunks/step01.sh` lifts only `dirent.h`, `lists.h` and the surgical `os-thread.cpp` edit. Include the aliases header when building this PR to cover the newlib `_READ_WRITE_RETURN_TYPE` fix (it is a real `xpack-development..smp` change).

3.2 The one code fix that is easy to get wrong (Step 13)

```
// C: include/cmsis-plus/rtos/port/os-inlines.h
inline clock_highres::timestamp_t clock_highres::cycles_since_tick (void) {
    uint32_t load = SysTick->LOAD, val = SysTick->VAL;
    if ((SCB->ICSR & SCB_ICSR_PENDSTSET_Msk) && (val > (load / 2)))
        return (load - val) + load;
    return load - val;
}
```

4. Phase B — SMP infrastructure (PR #2, Steps 14–23, collapsed into [step/14](#))

Why one blob. Unlike Part A, all ten steps' SMP code is interleaved inside the same `#if defined(OS_USE_SMP_SCHEDULER)` blocks in the same files. The whole set lands as **one commit** ([step/14](#), recipe [chunks/part-b.sh](#)) across the three repos, then the hardening fixes A–J are applied. **Every** change is inside the `OS_USE_SMP_SCHEDULER` guard, so the single-core binary is unchanged (proved by `unifdef`). The ten rows below are the conceptual sub-steps of that one commit; steps 15–23 do not exist as separate branches.

4.1 The SMP steps

Step	What	Why	File	Code / how
14	<code>port_cpu_id()</code>	identify the current core; TLS on the host, constant 0 on single-core Cortex-M	K: <code>src/rtos/os-core.cpp</code> , <code>os-thread.cpp</code> ; C: <code>os-inlines.h</code> ; P: <code>src/rtos/os-core.cpp</code>	P: <code>thread_local unsigned _this_cpu = 0; extern "C" __attribute__((noinline)) unsigned port_cpu_id(void) { asm volatile(":::"memory"); return _this_cpu; }</code>
15	Per-CPU scheduler lock <code>lock_state[OS_NCPU]</code>	one scheduler lock per core	K+P+C <code>src/rtos/os-core.cpp</code>	scalar → array; in P mask <code>SIGALRM</code> before reading <code>port_cpu_id()</code>
16	Recursive kernel lock <code>_smp_klock</code>	serialize kernel entry/exit across cores	<code>include/cmsis-plus/rtos/port/os-decls.h</code> , <code>os-c-decls.h</code>	<code>struct smp_klock_t { atomic lock; owner; count; }; _smp_klock_enter/exit</code> ; clear owner before the release store; <code>SMP_NO_OWNER 0xFFFFFFFFU</code>
17	Per-CPU interrupt tracking	know if the current core is in handler mode	P <code>os-decls.h</code>	<code>irq_set + _in_isr[OS_NCPU]</code> ; <code>in_handler_mode()</code> reads <code>_in_isr[port_cpu_id()]</code>
18	Per-CPU current thread <code>current_thread[OS_NCPU]</code>	each core has its own running thread	<code>os-sched.h</code>	array; <code>this_thread::thread()</code> reads <code>[port_cpu_id()]</code> with local IRQs masked
19	Per-CPU idle threads <code>os_idle_thread_core[OS_NCPU]</code>	every core needs an idle context	K+C+P <code>os-idle.cpp</code>	each core registers its private idle context at boot
20	CPU affinity mask	pin threads to cores; main thread to core 0	<code>os-thread.cpp</code>	<code>th_cpu_affinity</code> (default <code>0xFFFFFFFF</code>), <code>cpu_affinity()</code> API, main pinned to <code>1U<<0</code>
21	5-stage deferred publish/claim	hand a thread between cores without losing the stack pointer	K+C+P	claim (<code>to->stack_ptr=nullptr</code>) → spill → SP switch → publish (<code>from->stack_ptr=sp</code> ,

Step	What	Why	File	Code / how
				release) → restore; picker gate state_ != running && stack_ptr != nullptr
22	SMP ready-list picker	choose the highest-priority runnable thread for this core	os-core.cpp internal_switch_th reads	(affinity & (1<=cpu)) and published stack; else os_idle_thread_cor e[cpu]
23	POSIX host multiprocessing / IPI	model cores as pthreads; wake a core by signal	P src/ host_cpu.cpp , include/ host_cpu.hpp ; K weak port_smp_ipi	spawn OS_NCPU pthreads; per-CPU timer_create(SIGEV _THREAD_ID) ; IPI via pthread_kill(.., SIGRTMIN+1)

4.2 The files lifted for Part B

Repo	Files (whole, SMP versions)
K	include/cmsis-plus/rtos/os-c-decls.h , os-sched.h , os-thread.h ; src/rtos/os-c- wrapper.cpp , os-core.cpp , os-idle.cpp , os- main.cpp , os-thread.cpp
C	include/cmsis-plus/rtos/port/os-c-decls.h , os- decls.h , os-inlines.h ; src/rtos/os-core.cpp
P	include/cmsis-plus/rtos/port/os-c-decls.h , os- decls.h , os-inlines.h ; include/host_cpu.hpp , exception_handler.hpp , hw_result.hpp ; src/ host_cpu.cpp , free-store.cpp , board-contract.cpp , exception_handler.cpp , src/ rtos/os-core.cpp

4.3 The lift-time corrections (fixes A–J)

Fix	What	Why	File	Code
A/B	<code>struct timespec</code> <code>tp;</code> → <code>timespec</code> <code>tp;</code>	baseline gate is <code>-Werror=redundant-tags</code>	P <code>os-inlines.h</code>	drop the <code>struct</code> tag
C	Register new SMP sources additively	keep the xpack INTERFACE model, not smp's standalone <code>UOS_SMP_DIR</code>	P <code>CMakeLists.txt</code>	append to <code>target_sources(... INTERFACE)</code> ; do not build <code>board-contract.cpp</code> here
D	Register <code>host_cpu.cpp</code> , <code>free-store.cpp</code> , <code>exception_handler.cpp</code>	the refactored always-compiled <code>os-core.cpp</code> delegates to them	P <code>CMakeLists.txt</code>	<code>target_sources(... INTERFACE src/host_cpu.cpp src/free-store.cpp src/exception_handler.cpp)</code>
E	Drop redundant <code>port_cpu_id</code> decl; declare <code>port_smp_ipi</code>	<code>-Werror=missing-declarations</code>	P <code>host_cpu.cpp</code>	remove dup decl; add prior prototype for the strong override
F/G	Remove redundant <code>port_cpu_id</code> decl; weak <code>g_core_stage[]</code>	<code>-Werror=redundant-decls</code> ; symbol only defined by the SMP test board	P <code>exception_handler.cpp</code>	add a port-side weak definition of <code>g_core_stage[OS_NCPU]</code>
H	Wrap kernel <code>port_cpu_id</code> redundant decls	<code>-Werror=redundant-decls</code> only under a true SMP compile	K <code>os-core.cpp</code> , <code>os-thread.cpp</code>	<code>#pragma GCC diagnostic ignored "-Wredundant-decls"</code>
I/J	clang <code>-Weverything</code> hardening	destination is stricter than <code>smp</code>	P port headers + SMP <code>.cpp</code>	add <code>-Wc++98-compat-pedantic</code> / <code>-Wreserved-identifier</code> / <code>-Wunsafe-buffer-usage</code> ; suppression block before includes

4.4 The dual-core test that must ship with Part B (`native-smp`)

Part B's own gate (`verify-step.sh` Stage 4) is only a smoke build, and the runbook records that the original NCPU=2 command-line build was **fake** — the `-D` values became CMake cache variables, not compile definitions, so the code built single-core and the "SMP" run proved nothing. The only way SMP is truly on is a platform that sets the defines via `target_compile_definitions`. The `native-smp` platform does exactly that, so it belongs in PR #2:

What	Why	File	Code
Add the <code>native-smp</code> platform	a genuine dual-core host platform is the only real SMP test for Part B	<code>tests/platforms/native-smp/</code> (mirrors <code>native</code> ; links <code>micro-os-plus::iii-posix-arch</code> + <code>libucontext</code>)	<code>target_compile_definitions(... PRIVATE OS_INTEGER_RTOS_PORT_NCPU=2 OS_USE_SMP_SCHEDULER=1)</code>
Add its configs and actions	run it from xpm	<code>tests/package.json</code> (append-only)	<code>native-smp-cmake-gcc14-debug/release</code> ; <code>test-native-smp</code> ; introduce <code>test-smp-all</code> with this leg
Wire it into the harness	select the platform	<code>tests/cmake/tests-main.cmake</code> (additive)	platform chosen by <code>PLATFORM_NAME</code>

This is the change that surfaced kernel fix **H** (the `port_cpu_id` redundant-decl clash under a real SMP compile), already in `part-b.sh`. `2xcortex-m33` stays in PR #6 because it also needs the M33 port (Step 25) and the modular CMake (Step 26).

Gate for PR #2: pristine check + `unifdef -UOS_USE_SMP_SCHEDULER` single-core delta = 0 + 72/72 + the genuine `native-smp` dual-core suite (`rtos-apis-test`, `mutex-stress-test`, `cmsis-os-validator-test`).

5. Phase C1 — Port releases (PR #3, Step 24)

Why now. The releases publish the SMP port model that Part B just landed; they depend only on Part B, not on the new boards or tests, so they are their own small PR rather than being buried in the tests PR.

Step	What	Why	File / path	How
24	Port releases	publish the SMP port model (<code>posix-arch v1.1.0</code> , <code>cortexm v1.2.0</code>)	P/C <code>package.json</code> (<code>version</code> only)	<code>release-port.sh</code> edits <code>version</code> in Python and tags by hand — never <code>npm version</code> , which runs <code>postversion</code> and pushes

Gate for PR #3: the release commits touch only the `version` line; both tags exist locally only; the port branches still build.

6. Phase C2 — New cores/boards + modular CMake (PR #5, Steps 25–26)

Why here. These are **evolutions**: the M33 and RP2350 cores and the additive CMake that lets a new platform link the fat `::iii` without touching the existing targets. They need Part B (the SMP port contract) and Part 0 (the SoC code), and they unblock the tests in PR #6.

Step	What	Why	File / path	How
25	New cores/boards	M33 and RP2350 support	C <code>include-m33/</code> , <code>include-rp2350/</code> , <code>os-core-m33.cpp</code> , <code>os-core-rp2350.cpp</code> ; P <code>board-contract.cpp</code> , <code>free-store.cpp</code> , <code>exception_handler.*</code> ; K IPI in <code>thread::resume()</code>	add-only from <code>origin/smp</code>
26	Modular CMake (integrated, add-only)	build the new M33 target without breaking the fat <code>::iii</code>	C <code>CMakeLists.txt</code> (additive <code>micro-os-plus::cortexm-qemu-m33</code>); K <code>CMakeLists.txt</code> (additive harness sub-targets <code>micro-os-plus::iii-core</code> , <code>::iii-posix-io</code> , <code>::port-smp-decls</code> , <code>::test-support</code> + re-inclusion guard); <code>tests/platforms/2xcortex-m33/</code> ; <code>tests/device-qemu-cortexm-m33/</code> ; <code>tests/package.json</code> config + action	append the port target; new platform links the fat <code>::iii</code> ; self-contained device; <code>mps2-an521 --cpu cortex-m33 --smp 2</code>

Gate for PR #5: the original `xpm run test-all` (72) still green; the new M33 platform configures and builds.

7. Phase C3 — New test sources + add-only platforms (PR #6, Steps 27–28)

Why here. These are the **tests**: existing, proven `smp` suites promoted add-only, plus the AArch platforms. They need the cores/boards from PR #5.

Step	What	Why	File / path	How
27	New test sources	prove SMP on the new architectures	<code>tests/sources/fp-switch/</code> ; <code>tests/smp-support/</code> (dissolved <code>test_smp1/</code>); smp tests 0–5; AN505/AN521 linker scripts	<code>absorb-test-smpl.sh</code> ; append <code>set(ENABLE_FP_SWITCH_TEST true)</code> to <code>global-definitions.cmake</code>
28	Add-only platforms and harness wiring	run the suites on new targets	<code>tests/platforms/2xcortex-m33</code> , <code>cortexm-pico2</code> , <code>aarch32-rpi3b</code> , <code>aarch64-rpi3b</code> (the <code>native-smp</code> leg already landed in PR #2); <code>tests/cmake/tests-main.cmake</code> (additive); <code>tests/package.json</code>	<code>integrate-aarch-harness.sh</code> wires AArch32/64 platforms and dev-linked <code>aarch*-actions</code> in <code>package.json</code> (auto <code>xpm link</code> in <code>install</code> , <code>link-deps</code> , <code>link-deps-all</code>); extend <code>test-smp-all</code> with M33/aarch legs

Gate for PR #6: the original `xpm run test-all` (72) still green **and** `xpm run test-smp-all` green.

8. Phase D — Documentation and final merge (PR #7, Steps 29–30)

Step	What	Why	File	Code
29	Documentation	keep the runbook/PDF in step with the code	<code>docs/*.md</code> , <code>docs/*.pdf</code> , <code>docs/render-pdfs.sh</code>	<code>cd docs && ./render-pdfs.sh</code> <code>Implementation-SMP-Integration</code>
30	Final merge (local)	reconcile with upstream and drop dev tooling	all repos; remove <code>scripts/smp/</code>	merge <code>origin/xpack-development</code> ; restore <code>.github/</code> , <code>README</code> , <code>LICENSE</code> ; <code>finalize.sh</code> ; run both gates

Gate for PR #7: clean doc build; full ecosystem `test-all` + `test-smp-all`; `.github/README/LICENSE` byte-identical to baseline.

9. Per-step commands (the progressive loop)

One step = four commands. Nothing is pushed.

#	Command	What it does
1	<code>scripts/smp/new-step.sh NN</code>	fork <code>step/NN</code> from the most recent existing step branch (or the repo's <code>xpack-development</code>)
2	<code>scripts/smp/chunks/stepNN.sh</code> (or <code>chunks/part-b.sh</code> for the collapsed <code>step/14</code>)	apply the exact files + corrections for the step
3	<code>scripts/smp/verify-step.sh NN</code>	pristine check → <code>unifdef</code> (Part B / <code>step/14</code>) → link ports → 72/72 tests
4	<code>scripts/smp/advance-step.sh NN</code> <code>"message"</code>	commit the touched repos and tag <code>step-NN-green</code>

Drive Part A and the collapsed Part B with `scripts/smp/run-loop.sh 1 14`, then Part C with `scripts/smp/run-loop.sh 24 30`; finish with `full-verify.sh` (mandatory), then `finalize.sh`.

10. Opening the pull request (manual, by hand)

Nothing is automated up to this point. When the series is green locally, open the progressive PR yourself:

```
git -C "$WORK/micro-os-plus-iii" push origin step/30

gh pr create --repo micro-os-plus/micro-os-plus-iii \
  --base xpack-development --head step/30 \
  --title "SMP multi-core integration (progressive, bisectable series)" \
  --body-file docs/smp-integration/pull-request.md
```

For the stacked variant, open each against the previous head: #2 `--base step/13`, #3 `--base step/14`, #5 `--base step/24`, #6 `--base step/26`,

7 `--base step/28`. #4 (`part0-devices`) bases on the port branch from #3; it is

out of band, so #5's port side is rebased on it before review.

11. Review and bisect guide

Reviewer question	Where to look	What proves it
Did the test framework stay frozen?	<code>scripts/smp/check-pristine.sh</code>	existing <code>package.json</code> / <code>tests/**</code> / <code>.github/**</code> entries never modified or deleted
Is the single-core binary unchanged in Part B?	<code>verify-step.sh</code> Stage 1	<code>unifdef -UOS_USE_SMP_SCHEDULER</code> projection is identical to the previous step
Does every step build and pass?	<code>verify-step.sh NN</code>	72/72 (24 builds × 3 tests) with <code>-Werror</code>
Which commit introduced a regression?	<code>git bisect</code> over <code>step-NN-green</code> tags	each tag is a labelled green checkpoint
Is the full ecosystem green?	<code>full-verify.sh</code>	real <code>xpm install → link-deps → prepare → build → test</code> per config

11.1 Pass criteria per part

Part	Steps	Pristine rule	Extra gate
Part A	1–13	total freeze	72/72
Part B	14–23	total freeze	72/72 + <code>unifdef</code> invariant + <code>native-smp</code> dual-core suite
Part C	24–28	append-only	72/72 + <code>test-smp-all</code>
Part D	29–30	not gated (restores <code>.github/**</code>)	doc build + both suites

11.2 Files that carry several steps (lift carefully)

File	Steps	Watch for
<code>src/rtos/os-c-wrapper.cpp</code>	4, 9, 20	<code>os_thread_state_destroying</code> , SMP affinity <code>#if</code>
<code>src/rtos/os-thread.cpp</code>	1, 9, 14, 20	exported symbol (1), destroying/join (9), <code>port_cpu_id</code> /affinity (14, 20)
<code>include/cmsis-plus/rtos/os-thread.h</code>	1, 7, 9, 14, 20	errno hunk (7), destroying enum (9), affinity (14, 20)
<code>include/cmsis-plus/rtos/os-c-decls.h</code>	9, 10, 16	<code>destroying = 7</code> (9), <code>os_condvar_t</code> <code>void* clock</code> (10), <code>smp_klock</code> (16)
<code>src/rtos/os-core.cpp</code>	14, 15, 16, 18, 19, 22	almost entirely Part B, all under <code>#if</code>

12. Quick reference — what / why / file / code

Phase	What	Why	Files	Code (representative)
Part 0	Fold <code>devices</code> into arch repos	upstream has no seventh repo	<code>soc/*</code> , <code>drivers/*</code> , CMake shims	<code>git subtree add -P <path> <devices> split/<sub></code>
Part A	13 single-core fixes	correct and valuable without SMP; keep Part B pure	<code>include/**</code> , <code>src/**</code> , ports	<code>git checkout origin/smp -- <files> / sc-project.py / _edit.py</code>
Part B	SMP scheduler, locks, IPI	multi-core support	<code>os-core.cpp</code> , <code>os-sched.h</code> , port headers, <code>host_cpu.cpp</code>	inside <code>#if defined(OS_USE_SMP_SCHEDULER)</code> ; <code>_smp_klock</code> , <code>current_thread_[OS_NCPU]</code>
Part C	New targets and tests	add-only coverage for M33/RP2350/AArch	<code>tests/platforms/**</code> , <code>tests/sources/**</code>	append <code>test-smp-all</code> ; <code>target_compile_definitions(... OS_USE_SMP_SCHEDULER=1)</code>
Part D	Docs + final merge	land the series cleanly	<code>docs/**</code> , <code>.github/**</code>	<code>finalize.sh</code> ; open PR by hand

13. Appendix — Step 31: Dedicated Raspberry Pi Pico 2

(`cortexm-pico2`) Hardware Platform Wiring

Why: Configures complete standalone hardware testing infrastructure for physical Raspberry Pi Pico 2 silicon (RP2350 dual ARM Cortex-M33).

Step	What	Why	File / path	How
31	Dedicated Pico 2 hardware platform	Hardware testing on physical RP2350 silicon; secondary core SIO boot mailbox handshake	C <code>src/rtos/os-core-rp2350.cpp</code> , <code>test/boards/pico2/</code> , <code>test/pico2/</code> ; K <code>tests/platforms/cortexm-pico2/</code>	standalone hardware SWD / OpenOCD configuration; SIO FIFO wake-up sequence

The one rule that keeps the series trustworthy: never define `OS_USE_SMP_SCHEDULER` on the `cmake / xpm` command line (a cache variable does not become a compile flag). It must come from a platform's `target_compile_definitions`, and you must see it on the real compile line (`cmake --build <dir> -v`). Otherwise the "SMP" build is silently single-core.